Mohamed Amine Herch

Industrial Electronics Engineer | Robotics, ROS 2, Embedded Systems & Digital Twins

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Profile

Robotics engineer focused on ROS 2 navigation and multi-robot fleets. Integrated a 3-robot, 2-manufacturer AMR fleet (MiR, Omron) with Open-RMF and a Unity digital twin during a final-year project in Canada. Works in C++ and Python on Linux, with simulation experience in Gazebo and an embedded-systems background (STM32, ESP32).

Professional Experience

Final-Year Project: Heterogeneous AMR Fleet Integration & Digital Twin 2026

CNIMI Laboratory (UQTR) *Drummondville, Canada*

- Integrated 3 AMRs from 2 manufacturers (MiR250, MiR600, Omron LD-90) into one ROS 2 Jazzy / Open-RMF fleet.
- Bridged vendor interfaces (MiR REST/WebSocket, Omron ARCL) to Open-RMF so one dispatcher coordinates all robots.
- Built a Unity 6 digital twin (ROS-TCP-Connector) showing live robot positions and states to supervise the fleet in 3D.
- Validated simultaneous 3-robot missions with shared maps, lanes and traffic rules through repeated physical tests.

Keywords: *ROS 2 Jazzy, Open-RMF, Python (rclpy), Nav2, tf2, RViz2, REST API, WebSocket, TCP, Unity 6, digital twin, Linux*

Robotics Intern 2025

Rosenberger *Sousse, Tunisia*

- Developed autonomous navigation and SLAM simulations in ROS 2 and Gazebo to validate behavior before hardware tests.
- Built a mobile robot prototype (Raspberry Pi, LiDAR, DC motors with encoders) to test the navigation stack on real hardware.

Keywords: *ROS 2, Gazebo, SLAM, autonomous navigation, LiDAR, Raspberry Pi, DC motors, encoders*

Intern 2024

LEONI *Sousse, Tunisia*

- Gained exposure to industrial wiring harness production, assembly processes and quality control.

Keywords: *Manufacturing, automotive wiring harness, quality control, industrial processes*

Maker & Trainer 2024 – Present

FabLab ENISo *Sousse, Tunisia*

- Delivered 3 hands-on robotics and electronics workshops to groups of 12–14 students.
- Mentored student robotics and mechatronics projects from design through prototyping, integration and testing.

Keywords: *3D printing, rapid prototyping, laser engraving, CNC machining, technical training, mentoring*

Projects

- **ROS 2 Object Tracking (TurtleBot):** Built a ROS 2 object tracker on a TurtleBot, tested in simulation and on a real robot.
  Keywords: *ROS 2 Jazzy, Python, computer vision, multi-object tracking, SORT, Kalman filter, TurtleBot3*
- **SmartTime IoT:** Built an ESP8266 RFID attendance system publishing over MQTT (HiveMQ) to a live Node-RED dashboard.
  Keywords: *ESP8266, RFID, MQTT, HiveMQ, Node-RED, IoT*
- **Line-Following Robot:** Built a line-following robot using PID control and Pololu encoder feedback for trajectory tracking.
  Keywords: *PID control, closed-loop control, encoders, embedded systems*
- **Sign Language Recognition:** Developed a MATLAB gesture-recognition system using image processing and ML classification.
  Keywords: *MATLAB, image processing, machine learning, classification*

Technical Skills

ROS 2	ROS 2 Jazzy, Nav2, SLAM, tf2, RViz2, launch files, QoS tuning, ROS 1 to ROS 2 migration
Fleet & Integration	Open-RMF (fleet adapters, task dispatch, Traffic Editor), WebSocket (rosbridge), REST API, TCP
Programming	Python (rclpy), C/C++, Git, Linux (Ubuntu 24.04)
Simulation & Digital Twin	Gazebo, Unity 6 (ROS-TCP-Connector, URDF Importer)
Perception	LiDAR, camera-based multi-object tracking (SORT-style tracker on TurtleBot3)
Embedded & IoT	STM32, ESP32/ESP8266, Arduino, Raspberry Pi, MQTT, FPGA/VHDL
Tools	MATLAB, LaTeX, Altium, 3D printing

Education

Engineering Degree in Industrial Electronics, National Engineering School of Sousse (ENISo), Tunisia 2023 – 2026

Scientific Preparatory Cycle, Preparatory Institutes for Engineering Studies of Sfax and Monastir, Tunisia 2021 – 2023

Leadership

President 2024 – 2025

ENISO Team *ENISo, Sousse*

- Led a 150+ member student technical club and organized trainings in Arduino, SolidWorks, MATLAB, STM32 and ESP32.
- Coordinated major student events, including ESC V9.
- Under my presidency, the club won **Best Robotics Club** and **Best Robotics Event** in Tunisia (2024–2025).

Languages

Arabic (native) | French (professional) | English (professional)